

# Standard paper-and-pencil neuropsychology tests - Insight46 phase 1

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## Document details

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Categories of variables</b>    | Standard paper-and-pencil neuropsychology tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Summary of work undertaken</b> | <p>1. Data collected as described below.</p> <p>2. FNAME data cleaned as described below.</p> <p>3. Preclinical Alzheimer's Cognitive Composite (PACC) generated according to methods described below.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Names of source variables</b>  | <p>The, following, variables, relate, to, the, MMSE, and, were, all, extracted, from, XNAT, I, have, renamed, them, to, give, them, more, intuitive, names, I, have, provided, an, annotated, copy, of, the, XNAT, MMSE, form, at, the, end, of, this, document, in, order, to, identify, each, variable, id, ID, number), Ecomp, Ewn, Eyear, Eseason, Edate, Eday, Emonth, Ecountry, Ecity, Earea, Efloor, Ename, Eorientation, out, of, for, the, orientation, questions), Ereg_first1, Ereg_second1, Ereg_third1, Ereg_first2, Ereg_second2, Ereg_third2, Ereg_first3, Ereg_second3, Ereg_third3, Esubtract1, Esubtract2, Esubtract3, Esubtract4, Esubtract5, Esubtract_tot, Ebackwards, Ehighest, Erecall_first, Erecall_second, Erecall_third, Ewatch, Epen, Erepitition, Erighthand, Efolded, Elap, Ereading, Esentence, Edrawing, Es, The, following, variables, relate, to, the, Logical, Memory, Matrix, Reasoning, Digit-Symbol, Substitution, and, FNAME, tests, They, were, all, extracted, from, XNAT, I, have, renamed, them, to, give, them, more, intuitive, names, and, provided, annotated, copies, of, the, XNAT, forms, at, the, end, of, this, document, in, order, identify, each, variable, Licomp, Liwn, Lis, Ldcomp, Ldwn, Lt, Lds, Mcomp, Mwn, Ms, Wcomp, Wwn, Ws, Fcomp, Fwn, Finr1, Fior1, Finr2, Fior2, Finrt, Fiort, Fcrn, Fcro, Frecog, Fcrn30, Fcro30, Fmcn, Fmco, Ffnn, Ffno, Ftot</p>                                                                                                                                                                                                            |
| <b>Output variables</b>           | <p>mmse_comp_i46p1, mmse_wn_i46p1, mmse_year_i46p1, mmse_season_i46p1, mmse_date_i46p1, mmse_day_i46p1, mmse_month_i46p1, mmse_country_i46p1, mmse_city_i46p1, mmse_area_i46p1, mmse_floor_i46p1, mmse_name_i46p1, mmse_orientation_i46p1, mmse_reg_first1_i46p1, mmse_reg_second1_i46p1, mmse_reg_third1_i46p1, mmse_reg_first2_i46p1, mmse_reg_second2_i46p1, mmse_reg_third2_i46p1, mmse_subtract1_i46p1, mmse_subtract2_i46p1, mmse_subtract3_i46p1, mmse_subtract4_i46p1, mmse_subtract5_i46p1, mmse_subtract_tot_i46p1, mmse_backwards_i46p1, mmse_highest_i46p1, mmse_recall_first_i46p1, mmse_recall_second_i46p1, mmse_recall_third_i46p1, mmse_watch_i46p1, mmse_pen_i46p1, mmse_repetition_i46p1, mmse_righthand_i46p1, mmse_folded_i46p1, mmse_lap_i46p1, mmse_reading_i46p1, mmse_sentence_i46p1, mmse_drawing_i46p1, mmse_total_i46p1, mmse_zscore_i46p1, logmemimmcomp_i46p1, logmemimmtotal_i46p1, logmemimmzscore_i46p1, matrixcomp_i46p1, matrixtotal_i46p1, matrixzscore_i46p1, digsymsubcomp_i46p1, digsymsubnotcomp_i46p1, digsymsubtotal_i46p1, digsymsubzscore_i46p1, logmemdelcomp_i46p1, logmemdeltime_i46p1, logmemdelttotal_i46p1, logmemdelzscore_i46p1, fname_comp_i46p1, fname_wn_i46p1, fname_inr1_i46p1, fname_ior1_i46p1, fname_inr2_i46p1, fname_ior2_i46p1, fname_inrtotal_i46p1, fname_iortotal_i46p1, fname_crn_i46p1, fname_cro_i46p1, fname_recog_i46p1, fname_crn30_i46p1, fname_cro30_i46p1, fname_mcn_i46p1, fname_mco_i46p1, fname_namestotal_i46p1, fname_occstotal_i46p1, fname_total_i46p1, fname_nameszscore_i46p1, fname_occszscore_i46p1, fname_totalzscore_i46p1, pacc_i46p1</p> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Papers using these variables</b> | <p>1. Lu et al. (2019) Cognition at age 70: life course predictors and associations with brain pathologies, <i>Neurology</i>, in press</p> <p>2. Characterising cognition in later life and its relationship with biomarkers of Alzheimer's disease: a study of members of the National Survey of Health and Development (the British 1946 birth cohort). PhD thesis (Kirsty Lu)</p> <p>3. Blood and cerebrospinal fluid biomarkers in Alzheimer's disease – from clinical to preclinical cohorts. PhD thesis (Ashvini Keshavan).</p> <p>4. Plasma amyloid, tau and serum neurofilament light chain in Insight 46 – associations with cognition and brain imaging. AACC 2019 poster (Keshavan et al.)</p> |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Background to cognitive tests

Full details of these cognitive tests have been published in the following two papers: 1. Lane et al. (2017) Study protocol: Insight 46 – a neuroscience sub-study of the MRC National Survey of Health and Development, *BMC Neurology*, 17:75, doi: 10.1186/s12883-017-0846-x 2. Lu et al. (2019) Cognition at age 70: life course predictors and associations with brain pathologies, *Neurology*, in press A summary is provided below:

The **Mini Mental State Examination (MMSE)** is a 30-point composite screening tool for cognitive impairment which is widely used within clinical practice (Folstein, M. F., Folstein, S. E. and McHugh, P. R. (1975) "Mini-mental state." A practical method for grading the cognitive state of patients for the clinician., *Journal of Psychiatric Research*, 12(3), pp. 189–198. doi: 10.1016/0022-3956(75)90026-6). The item-by-item scores are provided in this dataset, as well as the total score.

**Digit-Symbol Substitution** from the Wechsler Adult Intelligence Scale-Revised is an index of executive function and psychomotor speed (Wechsler D. (1981b) *Wechsler Adult Intelligence Scale - Revised*. San Antonio, TX: The Psychological Corporation). The score is the number of items completed correctly within 90 seconds.

**Logical Memory IIa** from the Wechsler Memory Scale assesses free recall of a short story, which the participant is asked to recall immediately and after a delay of approximately 20 minutes (Wechsler D. (1987) *Wechsler Memory Scale - Revised: Manual*. San Antonio, TX: Psychological Corporation). The maximum score is 25. The exact delay duration is recorded so that it can be accounted for in analyses.

The **Face-Name test (FNAME-12)** assesses associative memory for face-name and face-occupation pairs (Papp, K. V et al. (2014) Development of a psychometrically equivalent short form of the Face-Name Associative Memory Exam for use along the early Alzheimer's disease trajectory, *Clin Neuropsychol*. 2014/05/13, 28(5), pp. 771–785. doi:10.1080/13854046.2014.911351). Two versions exist: FNAME-12A and FNAME12-B. This study used FNAME-12A. Participants are shown 12 unfamiliar face-name and face-occupation pairs (e.g. "Sarah, Reporter"), with 8 seconds to study each one. They are then presented with each face and asked to recall the associated name and occupation. This process is repeated with a second learning phase and a second recall test. After a ~10-minute delay they are again shown each face and asked to recall the names and occupations (the third recall test). After a ~30-minute delay participants are shown 12 sets of three faces and asked to identify each previously learned face from the two distractors (facial recognition) and to recall the name and occupation (the fourth recall test). If they cannot recall the name and/or occupation, they are asked to select the correct answer from three options comprising: the correct answer, a distractor (a name/occupation that belongs with a different face in the set), and a name/occupation that did not feature in the set. The summary outcomes are FN-N (total names recalled, max. 48), FN-O (total occupations recalled, max. 48) and FNAME-total (FN-N + FN-O, max. 96) – these outcomes are based on the four recall tests.

The **Matrix Reasoning** test assesses non-verbal reasoning, an aspect of fluid intelligence (Wechsler D. (1999) *The Wechsler abbreviated scale of intelligence*. San Antonio, TX: The Psychological Corporation). Participants are shown a matrix of geometric shapes and are required to select the missing piece from five options. There are 32 matrices, graded in difficulty, and the test is discontinued when participants make four consecutive errors (or four errors within five consecutive items), as specified in the manual. Note that the full test contains 35 items but the final 3 items are only to be administered to younger adults, as per the instructions in the manual, so were

not used in Insight 46.

## Data cleaning

One participant did not complete the full FNAME test, and therefore her total scores (Ffnn, Ffno, Ftot) automatically generated in XNAT were not valid. These total scores were changed to missing. A note to explain this was added to the Fwn variable for this participant.

On the MMSE, it was noted that 2 participants who failed to name all 3 items correctly on the first attempt (see question 11 in the MMSE form below) had missing data for their second attempt. This omission does not affect their MMSE score, since the score for the naming items is based solely on the first attempt. This is likely to be an administration error, as other versions of the MMSE do not require multiple attempts on this question. No other data-cleaning was required.

## Calculation of the Preclinical Alzheimer Cognitive Composite (PACC)

The four components of our version of the PACC were: MMSE total score, Logical Memory delayed recall score, Digit-Symbol Substitution score and FNAME-total. The four components were converted into z-scores based on the full Insight 46 sample, using the following Stata command:

```
egen newvar = std(oldvar)
```

The four z-scores were then averaged. A higher PACC score indicates better performance. Two participants did not complete the FNAME test and one did not complete the Digit-Symbol Substitution test. For these three participants, their PACC score was the average of the z-scores for the three tests they completed. Note that the original PACC (Donohue, M. C. et al. (2014) 'The preclinical Alzheimer cognitive composite: measuring amyloid-related decline.', JAMA neurology, 71(8), pp. 961–70. doi: 10.1001/jamaneurol.2014.803) contained the Free and Cued Selective Reminding Test (FCSRT) instead of FNAME. We replaced the FCSRT with FNAME to avoid potential overlap with a similar word-learning memory test administered to the NSHD cohort at multiple time-points throughout adulthood.

**MMSE XNAT form** - annotated in red with names of variables contained in this dataset

|                                                       |                                                                |             |           |
|-------------------------------------------------------|----------------------------------------------------------------|-------------|-----------|
| May I ask you some questions about your memory? Ecomp | 0 No                                                           |             |           |
|                                                       | 1 Yes                                                          |             |           |
| 2 Unable for health reasons                           |                                                                |             |           |
| 3 Unable, other _ _ _ _ _<br>_ _ _ _ _                |                                                                |             |           |
| If 0, 2 or 3 are ticked, specify reason Ewn           |                                                                |             |           |
| <b>ORIENTATION</b>                                    |                                                                |             |           |
| 1                                                     | What is the year? _ _ _ _<br>Eyear                             | 0 Incorrect | 1 Correct |
| 2                                                     | What is the season? _ _ _<br>_ _ _ Eseason                     | 0 Incorrect | 1 Correct |
| 3                                                     | What is the date? _ _ / _ _ /<br>_ _ _ _ (DD/MM/YYYY)<br>Edate | 0 Incorrect | 1 Correct |
| 4                                                     | What is the day of the week? _ _ _ _ _<br>Eday                 | 0 Incorrect | 1 Correct |
| 5                                                     | What is the month? _ _ _ _<br>_ _ Emonth                       | 0 Incorrect | 1 Correct |

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 6                                                      | Can you tell me where you are now? For instance what country are we in?<br>_ _ _ _ _ Ecountry                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0 Incorrect | 1 Correct |
| 7                                                      | What is the name of this city? _ _ _ _ _ Ecity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0 Incorrect | 1 Correct |
| 8                                                      | What area of the city are we in? Earea                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0 Incorrect | 1 Correct |
| 9                                                      | What floor of the building are we on? _ _ _ _ _ Efloor                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0 Incorrect | 1 Correct |
| 10                                                     | What is the name of this place? _ _ _ _ _ Ename                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0 Incorrect | 1 Correct |
| <b>REGISTRATION</b>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |           |
| 11                                                     | I am now going to name three objects. The three objects are:<br>APPLE, TABLE, PENNY.<br>Please repeat the name of these three objects back to me.<br>Record the FIRST responses (order of object recall does NOT matter):                                                                                                                                                                                                                                                                                                           |             |           |
| First object named? _ _ _ _ _<br>_ _ Ereg_first1       | 0 Incorrect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 Correct   |           |
| Second object named? _ _ _ _ _<br>_ _ _ _ Ereg_second1 | 0 Incorrect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 Correct   |           |
| Third object named? _ _ _ _ _<br>_ _ _ _ Ereg_third1   | 0 Incorrect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 Correct   |           |
|                                                        | If all three objects are repeated correctly (in any order) go to Q12a<br>If the participant does not repeat all three words exactly then allow them two further attempts but DO NOT change their first responses or scores.<br>Attempt 2: I'm going to repeat once more the three objects, APPLE, TABLE, PENNY. Please say them back to me.<br>Ereg_first2 Ereg_second2 Ereg_third2<br>Attempt 3: Can we try one more time? The names are, APPLE, TABLE, PENNY. Please say them back to me.<br>Ereg_first3 Ereg_second3 Ereg_third3 |             |           |
| <b>ATTENTION AND CALCULATION</b>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |           |

|                                                                                                                                                                                                                                                                                                                  |                                                                                                                              |                                                                                                         |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------|
| 12a                                                                                                                                                                                                                                                                                                              | Please take seven away from one hundred (Answer: 93)<br>Answer 1: __ _ Esubtract1                                            | 0 Incorrect                                                                                             | 1 Correct |
| Now I'm going to ask you to take seven away from what you have left over, and then keep taking seven away until I stop you. (Answers: 86, 79, 72 and 65. If the participant gets an answer wrong but then takes seven away correctly from that answer you should score as correct).<br>Answer 2: __ _ Esubtract2 | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
| Answer 3: __ _ Esubtract3                                                                                                                                                                                                                                                                                        | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
| Answer 4: __ _ Esubtract4                                                                                                                                                                                                                                                                                        | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
| Answer 5: __ _ Esubtract5                                                                                                                                                                                                                                                                                        | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
|                                                                                                                                                                                                                                                                                                                  | If the participant scores 5/5 then go to Q13.<br>If the participant scores less than 5/5 for 12a then complete question 12b. | Total score: _ /5<br>Esubtract_tot                                                                      |           |
| 12b                                                                                                                                                                                                                                                                                                              | Please spell "WORLD" backwards (Answer: D L R O W)                                                                           | TOTAL number of letters in correct order _ /5<br>Ebackwards                                             |           |
|                                                                                                                                                                                                                                                                                                                  | Highest score for subtraction or backwards spelling tests                                                                    | _ /5 Ehighest (note that this variable is only entered for participants who were administered item 12b) |           |
| <b>RECALL</b>                                                                                                                                                                                                                                                                                                    |                                                                                                                              |                                                                                                         |           |
| 13                                                                                                                                                                                                                                                                                                               | Please name the three objects that I mentioned to you earlier?                                                               |                                                                                                         |           |
| First object named __ _ _ _<br>__ _ Erecall_first                                                                                                                                                                                                                                                                | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
| Second object named __ _ _<br>__ _ Erecall_second                                                                                                                                                                                                                                                                | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
| Third object named __ _ _ _<br>__ _ Erecall_third                                                                                                                                                                                                                                                                | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
| <b>NAMING</b>                                                                                                                                                                                                                                                                                                    |                                                                                                                              |                                                                                                         |           |
| 14                                                                                                                                                                                                                                                                                                               | What is this called? (show watch) __ _ _ _ _ Ewatch<br>What is this called? (show pen) __ _ _ _ _ Epen                       | 0 Incorrect                                                                                             | 1 Correct |
| 15                                                                                                                                                                                                                                                                                                               | 0 Incorrect                                                                                                                  | 1 Correct                                                                                               |           |
| <b>REPETITION</b>                                                                                                                                                                                                                                                                                                |                                                                                                                              |                                                                                                         |           |

|                                                                               |                                                                                                                                                                                                                                                                                               |             |           |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 16                                                                            | I am going to say a sentence to you and I would like you to repeat it after I have said it. The sentence is: "NO IFS, ANDS OR BUTS."<br>Did the participant answer correctly? Do not repeat the sentence. If necessary say: I'm sorry, I'm only allowed to say the sentence once. Erepetition | 0 Incorrect | 1 Correct |
| <b>THREE STAGE COMMAND</b>                                                    |                                                                                                                                                                                                                                                                                               |             |           |
| 17                                                                            | Please take this paper with your right hand, fold it in half and put it on your lap. Do not repeat the sentence. If necessary say: I'm sorry, I'm only allowed to read that out once. Did the participant take the paper in their right hand? Erighthand                                      | 0 No        | 1 Yes     |
| Did the participant fold the paper in half or quarter (both allowed)? Efolded | 0 No                                                                                                                                                                                                                                                                                          | 1 Yes       |           |
| Did the participant put the paper on their lap? Elap                          | 0 No                                                                                                                                                                                                                                                                                          | 1 Yes       |           |
| <b>READING</b>                                                                |                                                                                                                                                                                                                                                                                               |             |           |
| 18                                                                            | Now give the sheet called "MMSE CLOSE YOUR EYES" to the participant and say: Please read the sentence at the top of this sheet and do what it says. Did the participant close their eyes? Ereading                                                                                            | 0 No        | 1 Yes     |
| <b>WRITING</b>                                                                |                                                                                                                                                                                                                                                                                               |             |           |
| 19                                                                            | Please write a sentence in the space here. Indicate the space under "Write a sentence". Did the participant write a sentence? Esentence                                                                                                                                                       | 0 No        | 1 Yes     |
| 20                                                                            | Please copy this drawing. Indicate the space to the right of the design. Did the participant copy the drawing correctly? Edrawing                                                                                                                                                             | 0 No        | 1 Yes     |
| Score                                                                         | __ /30 Es                                                                                                                                                                                                                                                                                     |             |           |

**Logical Memory Immediate Recall XNAT form** - annotated in red with names of variables contained in this dataset

|                                                        |      |
|--------------------------------------------------------|------|
| WMS Logical Memory – Immediate Recall completed Licomp | 0 No |
|--------------------------------------------------------|------|

|                                              |            |
|----------------------------------------------|------------|
| 1 Yes                                        |            |
| 2 Unable for health reasons                  |            |
| 3 Unable, other (Specify)_____               |            |
| If 0, 2 or 3 are ticked, specify reason Liwn |            |
| Total number of story units recalled:        | __ /25 Lis |

**Logical Memory Delayed Recall XNAT form** - annotated in red with names of variables contained in this dataset

|                                               |                 |
|-----------------------------------------------|-----------------|
| WMS Logical Memory – Delayed completed Ldcomp | 0 No            |
| 1 Yes                                         |                 |
| 2 Unable for health reasons                   |                 |
| 3 Unable, other (Specify)_____ Ldwn           |                 |
| If 0, 2 or 3 are ticked, specify reason Ldwn  |                 |
| Total number of story units recalled:         | __ /25 Lds      |
| Time elapsed since Logical Memory-Immediate:  | __ (minutes) Lt |

**Matrix Reasoning XNAT form** - annotated in red with names of variables contained in this dataset

|                                             |           |
|---------------------------------------------|-----------|
| WASI Matrix Reasoning completed Mcomp       | 0 No      |
| 1 Yes                                       |           |
| 2 Unable for health reasons                 |           |
| 3 Unable, other (Specify)_____ Mwn          |           |
| If 0, 2 or 3 are ticked, specify reason Mwn |           |
| Total correct:                              | __ /32 Ms |

**Digit-Symbol XNAT form** - annotated in red with names of variables contained in this dataset

|                                                     |           |
|-----------------------------------------------------|-----------|
| WAIS Digit Symbol Substitution Test completed Wcomp | 0 No      |
| 1 Yes                                               |           |
| 2 Unable for health reasons                         |           |
| 3 Unable, other (Specify)_____ Wwn                  |           |
| If 0, 2 or 3 are ticked, specify reason Wwn         |           |
| Total correct:                                      | __ /93 Ws |

**FNAME XNAT form** - annotated in red with names of variables contained in this dataset

|                                                   |                  |
|---------------------------------------------------|------------------|
| Face Name Associative Memory Exam completed Fcomp | 0 No             |
| 1 Yes                                             |                  |
| 2 Unable for health reasons                       |                  |
| 3 Unable, other (Specify)_____ Fwn                |                  |
| If 0, 2 or 3 are ticked, specify reason Fwn       |                  |
| <b>Learning Trial 1</b>                           |                  |
| INR1: Finr1<br>IOR1: Fior1                        | __ /12<br>__ /12 |
| <b>Learning Trial 2</b>                           |                  |

|                  |        |
|------------------|--------|
| INR2: Finr2      | __ /12 |
| IOR2: Fior2      | __ /12 |
| Total INR: Finrt | __ /24 |
| Total IOR: Fiort | __ /24 |

| Cued Name and Occupation Recall |        |
|---------------------------------|--------|
| CRN: Fcrn                       | __ /12 |
| CRO: Fcro                       | __ /12 |

| Delayed Cued Recall of Names and Occupations |        |
|----------------------------------------------|--------|
| Facial Recognition Frecog                    | __ /12 |
| CRN 30: Fcrn30                               | __ /12 |
| CRO30: Fcro30                                | __ /12 |
| MCN: Fmcn                                    | __ /12 |
| MCO: Fmco                                    | __ /12 |
| <b>Totals</b>                                |        |
| <b>FN-N</b> (Total INR + CRN + CRN30) Ffnn   | __ /48 |
| <b>FN-O</b> (Total IOR + CRO + CRO30 Ffno    | __ /48 |
| <b>Total FNAME Score</b> (CN-N + FN-O) Ftot  | __ /96 |

## Variables in this document

### 74 high-confidence matches

| Variable                                                                                                    | Label                                                                                     | How it was found |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------|
| <b>Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Associative episodic memory [60429]</b> |                                                                                           |                  |
| <a href="#">fname_comp_i46p1</a>                                                                            | Face-Name test (FNAME): was it completed? - Insight46 phase 1                             | topsheet field   |
| <a href="#">fname_crn30_i46p1</a>                                                                           | Face-Name test (FNAME): CRN30 Cued Name Retrieval long delay - Insight46 phase 1          | topsheet field   |
| <a href="#">fname_crn_i46p1</a>                                                                             | Face-Name test (FNAME): CRN Cued Recall Names short delay - Insight46 phase 1             | topsheet field   |
| <a href="#">fname_cro30_i46p1</a>                                                                           | Face-Name test (FNAME): CRO30 Cued Occupation Retrieval long delay - Insight46 phase 1    | topsheet field   |
| <a href="#">fname_cro_i46p1</a>                                                                             | Face-Name test (FNAME): CRO Cued Recall Occupations short delay - Insight46 phase 1       | topsheet field   |
| <a href="#">fname_inr1_i46p1</a>                                                                            | Face-Name test (FNAME): INR1 Names Learning Trial 1 - Insight46 phase 1                   | topsheet field   |
| <a href="#">fname_inr2_i46p1</a>                                                                            | Face-Name test (FNAME): INR2 Names Learning Trial 2 - Insight46 phase 1                   | topsheet field   |
| <a href="#">fname_inrtotal_i46p1</a>                                                                        | Face-Name test (FNAME): INR Total Names Learning Trials 1 and 2 - Insight46 phase 1       | topsheet field   |
| <a href="#">fname_ior1_i46p1</a>                                                                            | Face-Name test (FNAME): IOR1 Occupations Learning Trial 1 - Insight46 phase 1             | topsheet field   |
| <a href="#">fname_ior2_i46p1</a>                                                                            | Face-Name test (FNAME): IOR2 Occupations Learning Trial 2 - Insight46 phase 1             | topsheet field   |
| <a href="#">fname_iortotal_i46p1</a>                                                                        | Face-Name test (FNAME): IOR Total Occupations Learning Trials 1 and 2 - Insight46 phase 1 | topsheet field   |
| <a href="#">fname_mcn_i46p1</a>                                                                             | Face-Name test (FNAME): MCN Multiple Choice Names - Insight46 phase 1                     | topsheet field   |
| <a href="#">fname_mco_i46p1</a>                                                                             | Face-Name test (FNAME): MCN Multiple Choice Occupations - Insight46 phase 1               | topsheet field   |



| Variable                                                                                                         | Label                                                                                                                     | How it was found |
|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------|
| <a href="#">fname_namestotal_i46p1</a>                                                                           | Face-Name test (FNAME): FN-N Names Total Score - Insight46 phase 1                                                        | topsheet field   |
| <a href="#">fname_nameszscore_i46p1</a>                                                                          | Face-Name test (FNAME): Z-score of FN-N total names score - Insight46 phase 1                                             | topsheet field   |
| <a href="#">fname_occstotal_i46p1</a>                                                                            | Face-Name test (FNAME): FN-O Occupations Total Score - Insight46 phase 1                                                  | topsheet field   |
| <a href="#">fname_occszscore_i46p1</a>                                                                           | Face-Name test (FNAME): Z-score of FN-O total occupations score - Insight46 phase 1                                       | topsheet field   |
| <a href="#">fname_recog_i46p1</a>                                                                                | Face-Name test (FNAME): Facial Recognition - Insight46 phase 1                                                            | topsheet field   |
| <a href="#">fname_total_i46p1</a>                                                                                | Face-Name test (FNAME): Total Score - Insight46 phase 1                                                                   | topsheet field   |
| <a href="#">fname_totalzscore_i46p1</a>                                                                          | Face-Name test (FNAME): Z-score of total score - Insight46 phase 1                                                        | topsheet field   |
| <b>Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Fluid intelligence and reasoning [60424]</b> |                                                                                                                           |                  |
| <a href="#">matrixcomp_i46p1</a>                                                                                 | Matrix Reasoning: was it completed? - Insight46 phase 1                                                                   | topsheet field   |
| <a href="#">matrixtotal_i46p1</a>                                                                                | Matrix Reasoning: score - Insight46 phase 1                                                                               | topsheet field   |
| <a href="#">matrixzscore_i46p1</a>                                                                               | Matrix Reasoning: Z-score - Insight46 phase 1                                                                             | topsheet field   |
| <b>Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Logical memory [60430]</b>                   |                                                                                                                           |                  |
| <a href="#">logmemdelcomp_i46p1</a>                                                                              | WMS-R Logical Memory Delayed: was it completed? - Insight46 phase 1                                                       | topsheet field   |
| <a href="#">logmemdeltime_i46p1</a>                                                                              | WMS-R Logical Memory: time elapsed for delay (min) - Insight46 phase 1                                                    | topsheet field   |
| <a href="#">logmemdeltotal_i46p1</a>                                                                             | WMS-R Logical Memory Delayed: score - Insight46 phase 1                                                                   | topsheet field   |
| <a href="#">logmemdelzscore_i46p1</a>                                                                            | WMS-R Logical Memory Delayed: Z-score - Insight46 phase 1                                                                 | topsheet field   |
| <a href="#">logmemimmcomp_i46p1</a>                                                                              | WMS-R Logical Memory Immediate: was it completed? - Insight46 phase 1                                                     | topsheet field   |
| <a href="#">logmemimmtotal_i46p1</a>                                                                             | WMS-R Logical Memory Immediate: score - Insight46 phase 1                                                                 | topsheet field   |
| <a href="#">logmemimmzscore_i46p1</a>                                                                            | WMS-R Logical Memory Immediate: Z-score - Insight46 phase 1                                                               | topsheet field   |
| <b>Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Mini-mental state examination [60422]</b>    |                                                                                                                           |                  |
| <a href="#">mmse_area_i46p1</a>                                                                                  | MMSE: orientation - what area of the city are we in? - Insight46 phase 1                                                  | topsheet field   |
| <a href="#">mmse_backwards_i46p1</a>                                                                             | MMSE: attention and calculation - WORLD backwards (out of 5), administered if subtraction not perfect - Insight46 phase 1 | topsheet field   |
| <a href="#">mmse_city_i46p1</a>                                                                                  | MMSE: orientation - what is the name of this city? - Insight46 phase 1                                                    | topsheet field   |
| <a href="#">mmse_comp_i46p1</a>                                                                                  | MMSE: May I ask you some questions about your memory? - Insight46 phase 1                                                 | topsheet field   |
| <a href="#">mmse_country_i46p1</a>                                                                               | MMSE: orientation - Can you tell me where you are now, for instance what country are we in? - Insight46 phase 1           | topsheet field   |
| <a href="#">mmse_date_i46p1</a>                                                                                  | MMSE: orientation - what is the date - Insight46 phase 1                                                                  | topsheet field   |
| <a href="#">mmse_day_i46p1</a>                                                                                   | MMSE: orientation - what is the day of the week - Insight46 phase 1                                                       | topsheet field   |
| <a href="#">mmse_drawing_i46p1</a>                                                                               | MMSE: copy drawing of overlapping pentagons - Insight46 phase 1                                                           | topsheet field   |

| Variable                 | Label                                                                                                                                    | How it was found |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| mmse_floor_i46p1         | MMSE: orientation - what floor of the building are we on? - Insight46 phase 1                                                            | topsheet field   |
| mmse_folded_i46p1        | MMSE: command - fold paper - Insight46 phase 1                                                                                           | topsheet field   |
| mmse_highest_i46p1       | MMSE: attention and calculation - highest out of subtraction or WORLD backwards if both were administered (out of 5) - Insight46 phase 1 | topsheet field   |
| mmse_lap_i46p1           | MMSE: command - put paper on lap - Insight46 phase 1                                                                                     | topsheet field   |
| mmse_month_i46p1         | MMSE: orientation - what is the month - Insight46 phase 1                                                                                | topsheet field   |
| mmse_name_i46p1          | MMSE: orientation - what is the name of this place - Insight46 phase 1                                                                   | topsheet field   |
| mmse_orientation_i46p1   | MMSE: orientation total out of 10 - Insight46 phase 1                                                                                    | topsheet field   |
| mmse_pen_i46p1           | MMSE: naming - pen - Insight46 phase 1                                                                                                   | topsheet field   |
| mmse_reading_i46p1       | MMSE: reading (close your eyes) - Insight46 phase 1                                                                                      | topsheet field   |
| mmse_recall_first_i46p1  | MMSE: recall - first object - Insight46 phase 1                                                                                          | topsheet field   |
| mmse_recall_second_i46p1 | MMSE: recall - second object - Insight46 phase 1                                                                                         | topsheet field   |
| mmse_recall_third_i46p1  | MMSE: recall - third object - Insight46 phase 1                                                                                          | topsheet field   |
| mmse_reg_first1_i46p1    | MMSE: registration - first object attempt 1 - Insight46 phase 1                                                                          | topsheet field   |
| mmse_reg_first2_i46p1    | MMSE: registration - first object attempt 2 - Insight46 phase 1                                                                          | topsheet field   |
| mmse_reg_second1_i46p1   | MMSE: registration - second object attempt 1 - Insight46 phase 1                                                                         | topsheet field   |
| mmse_reg_second2_i46p1   | MMSE: registration - second object attempt 2 - Insight46 phase 1                                                                         | topsheet field   |
| mmse_reg_third1_i46p1    | MMSE: registration - third object attempt 1 - Insight46 phase 1                                                                          | topsheet field   |
| mmse_reg_third2_i46p1    | MMSE: registration - third object attempt 2 - Insight46 phase 1                                                                          | topsheet field   |
| mmse_repetition_i46p1    | MMSE: repetition - Insight46 phase 1                                                                                                     | topsheet field   |
| mmse_righthand_i46p1     | MMSE: command - take paper in right hand - Insight46 phase 1                                                                             | topsheet field   |
| mmse_season_i46p1        | MMSE: orientation - what is the season - Insight46 phase 1                                                                               | topsheet field   |
| mmse_sentence_i46p1      | MMSE: write a sentence - Insight46 phase 1                                                                                               | topsheet field   |
| mmse_subtract1_i46p1     | MMSE: attention and calculation - subtraction 1 - Insight46 phase 1                                                                      | topsheet field   |
| mmse_subtract2_i46p1     | MMSE: attention and calculation - subtraction 2 - Insight46 phase 1                                                                      | topsheet field   |
| mmse_subtract3_i46p1     | MMSE: attention and calculation - subtraction 3 - Insight46 phase 1                                                                      | topsheet field   |

| Variable                                                                                                                         | Label                                                                              | How it was found |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------|
| mmse_subtract4_i46p1                                                                                                             | MMSE: attention and calculation - subtraction 4 - Insight46 phase 1                | topsheet field   |
| mmse_subtract5_i46p1                                                                                                             | MMSE: attention and calculation - subtraction 5 - Insight46 phase 1                | topsheet field   |
| mmse_subtract_tot_i46p1                                                                                                          | MMSE: attention and calculation - subtraction total (out of 5) - Insight46 phase 1 | topsheet field   |
| mmse_total_i46p1                                                                                                                 | MMSE: score - Insight46 phase 1                                                    | topsheet field   |
| mmse_watch_i46p1                                                                                                                 | MMSE: naming - watch - Insight46 phase 1                                           | topsheet field   |
| mmse_year_i46p1                                                                                                                  | MMSE: orientation - what is the year - Insight46 phase 1                           | topsheet field   |
| mmse_zscore_i46p1                                                                                                                | MMSE: Z-score - Insight46 phase 1                                                  | topsheet field   |
| <b>Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Preclinical Alzheimer Cognitive Composite (PACC) [60432]</b> |                                                                                    |                  |
| pacc_i46p1                                                                                                                       | Preclinical Alzheimer Cognitive Composite (PACC) - Insight46 phase 1               | topsheet field   |
| <b>Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Symbol digit substitution [60425]</b>                        |                                                                                    |                  |
| digsymsubcomp_i46p1                                                                                                              | WAIS Digit Symbol Substitution Test: was it completed? - Insight46 phase 1         | topsheet field   |
| digsymsubtotal_i46p1                                                                                                             | WAIS Digit Symbol Substitution Test: score - Insight46 phase 1                     | topsheet field   |
| digsymsubzscore_i46p1                                                                                                            | WAIS Digit Symbol Substitution Test: Z-score - Insight46 phase 1                   | topsheet field   |

## 1 medium-confidence match

| Variable                                                                                                                         | Label                                                       | How it was found |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------|
| <b>Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]</b> |                                                             |                  |
| blood                                                                                                                            | Have you had anaemia in the last 10 years - at age 43 years | prose scan       |

## 2 low-confidence matches

| Variable                                                                                                                                  | Label                                                                                                                                                  | How it was found |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Clinical data [51] › Physical measures [110] › Hand examination [1106]</b>                                                             |                                                                                                                                                        |                  |
| hand                                                                                                                                      | Now I would like to examine your hands for any bumps or swellings. Would you be willing for me to examine your hands? - first record - at age 53 years | prose scan       |
| <b>Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]</b> |                                                                                                                                                        |                  |
| back                                                                                                                                      | In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years                                                               | prose scan       |